STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 14

RANDOM VARIABLES

JOINT CONTINUOUS RANDOM VARIABLES

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

JOINT DENSITY FUNCTIONS

CamScanner

JOINT DENSITY FUNCTIONS

If X and Y are continuous random variables. Then the function $f(x,y)$ is said to be a joint density function if it satisfy the following properties.

1- $f(x,y) \geq 0$ for all (x,y)

2- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

① MARGINAL PROBABILITY FUNCTIONS

Marginal probability functions of continuous Random variables X and Y are defined as

Marginal Prob. fn for X

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

Marginal Prob fn for Y

$$h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

② CONDITIONAL DISTRIBUTIONS

If X and Y are continuous RVS than distributions are given by

$$f(x/y) = \frac{f(x,y)}{h(y)}$$

also $$f(y/x) = \frac{f(x,y)}{g(x)}$$

③ INDEPENDENCE OF VARIABLES

If X and Y are continuous random variables than X and Y will be independent of each other iff

$$f(x,y) = g(x).h(y).$$

④ JOINT DISTRIBUTION FUNCTION

If X and Y are continuous random variables then the joint distribution function $F(x,y)$ is defined as

$$F(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(x,y)\,dx\,dy.$$

⑤ SOME IMPORTANT FORMULAS

1- $P(a<x<b,\ c<Y<d) = \int_a^b \int_c^d f(x,y)\,dy\,dx$

2- $P(a<x<b) = \int_a^b g(x)\,dx$

3- $P(c<Y<d) = \int_c^d h(y)\,dy$

4- $P(a<x<b/Y=c) = \int_a^b f(x/y)|_{y=c}\,dx$

5- $P(c<Y<d/X=a) = \int_c^d f(y/x)|_{x=a}\,dy.$

6- $P(a\le x\le b \mid c\le Y\le d) = \dfrac{\int_c^d\int_a^b f(x,y)\,dx\,dy}{\int_c^d h(y)\,dy}$

7- $P(c\le Y\le d \mid a\le x\le b) = \dfrac{\int_c^d\int_a^b f(x,y)\,dx\,dy}{\int_a^b g(x)\,dx}$

CamScanner

EXAMPLES

EXAMPLE - ① Suppose the random variables x and y have the joint density function.

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) verify $f(x,y)$ is a joint density function.

(b) Find $P(0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2})$

(c) Marginal density functions

(d) conditional density functions $f(x/y)$ and $f(y/x)$

(e) Are x and Y independent.

SOLUTION (a) verification

WKT $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$$\text{LHS} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_0^1\int_0^1 \frac{2}{5}(2x+3y)\,dx\,dy = 1 = \text{RHS}$$

(b) $P(0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2})$

$$= \int_0^{\frac{1}{2}}\int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5}(2x+3y)\,dy\,dx = \int_{\frac{1}{4}}^{\frac{1}{2}}\int_0^{\frac{1}{2}} \frac{2}{5}(2x+3y)\,dx\,dy = \frac{13}{160}$$

(c) Marginal density functions

$$g(x) = \int_0^1 \frac{2}{5}(2x+3y)\,dy = \frac{4x+3}{5}$$

$$h(y) = \int_0^1 \frac{2}{5}(2x+3y)\,dx = \frac{2}{5}(1+3y)$$

(d) conditional density functions

$$f(x/y) = \frac{f(x,y)}{h(y)} = \frac{\cancel{2}}{\cancel{5}}(2x+3y) \times \frac{\cancel{5}}{\cancel{2}(1+3y)} = \frac{2x+3y}{1+3y}$$

CamScanner

Also $f(y/x) = \frac{f(x,y)}{g(x)} = \frac{2}{5}(2x+3y) \times \frac{5}{4x+3} = \frac{2(2x+3y)}{4x+3}$

(e) Independence.

check $f(x,y) = g(x).h(y)$

$$\frac{2}{5}(2x+3y) \neq \frac{(4x+3)}{5} \times \frac{2}{5}(1+3y)$$

So x and y are not independent.

EXAMPLE ② The joint density function of the random variables x and y is given by

$$f(x,y) = \begin{cases} c(2x+y), & 2<x<6,\ 0<y<5 \\ 0, & \text{otherwise.} \end{cases}$$

Find the following

(a) the value of 'c'.

(b) $P(x<4, y>2)$

(c) Marginal density functions of x and Y.

(d) conditional density functions $f(x/y)$ and $f(y/x)$

(e) Are x and y independent.

SOLUTION (a) the value of 'c'.

WKT $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

or $\int_{0}^{5}\int_{2}^{6} c(2x+y)\,dx\,dy = 1 \Rightarrow c = \frac{1}{210}$

(b) $P(x<4, y>2)$

$$= \int_{2}^{4}\int_{2}^{5} \frac{1}{210}(2x+y)\,dy\,dx = \frac{57}{210}$$

(c) Marginal density functions

$$g(x) = \int_{0}^{5} \frac{1}{210}(2x+y)\,dy = \frac{4x+5}{84}$$

and $h(y) = \int_2^6 \frac{1}{210}(2x+y)\,dx = \frac{16+2y}{105}$

(d) conditional density functions.

$$f(x/y) = \frac{f(x,y)}{h(y)} = \frac{1}{210}(2x+y) \times \frac{105}{16+2y} = \frac{2x+y}{2(2y+16)}$$

$$f(y/x) = \frac{f(x,y)}{g(x)} = \frac{1}{210}(2x+y) \times \frac{84}{4x+5} = \frac{2(2x+y)}{5(4x+5)}$$

(e) independence.

check $f(x,y) = g(x).h(y)$

$$\frac{1}{210}(2x+y) \neq \frac{4x+5}{84} \times \frac{16+2y}{105}$$

so x and y are not independent.

EXAMPLE-③ Given the joint density function.

$$f(x,y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0<x<2,\ 0<y<1 \\ 0, & \text{elsewhere} \end{cases}$$

Find (a) $g(x)$, (b) $h(y)$, (c) $f(x/y)$ and (d) $P(\frac{1}{4}<X<\frac{1}{2} \mid Y=\frac{1}{3})$.
(e) Are x and y independent.

SOLUTION Now

(a) $g(x) = \int_0^1 \frac{x(1+3y^2)}{4}\,dy = \frac{x}{2}$

(b) $h(y) = \int_0^2 \frac{x(1+3y^2)}{4}\,dx = \frac{1+3y^2}{2}$

(c) $f(x/y) = \frac{f(x,y)}{h(y)} = \frac{x}{4}(1+3y^2) \cdot \frac{2}{(1+3y^2)} = \frac{x}{2}$

(d) $P(\frac{1}{4}<X<\frac{1}{2} \mid Y=\frac{1}{3}) = \int_{1/4}^{1/2} f(x/y)\big|_{y=\frac{1}{3}}\,dx = \int_{1/4}^{1/2} \frac{x}{2}\,dx = \frac{3}{64}$

(e) check $f(x,y) = g(x).h(y)$

$$\frac{x}{4}(1+3y^2) = \frac{x}{2} \cdot \frac{(1+3y^2)}{2}$$

$$\frac{x(1+3y^2)}{4} = \frac{x(1+3y^2)}{4}$$

So x and y are independent variables.

(*) Note that here $f(x/y) = g(x)$ so independent.

JOINT PDFS INVOLVING GRAPHS

CamScanner

EXAMPLES

EXAMPLE① Suppose two random variables x and y have joint p.d.f

$$f(x,y) = \begin{cases} 4xy, & 0<x<1,\ 0<y<1 \\ 0, & \text{elsewhere.} \end{cases}$$

Find (a) $P(0 \le x \le \frac{1}{2} \text{ and } \frac{1}{4} \le y \le \frac{1}{2})$

(b) $P(x<y)$

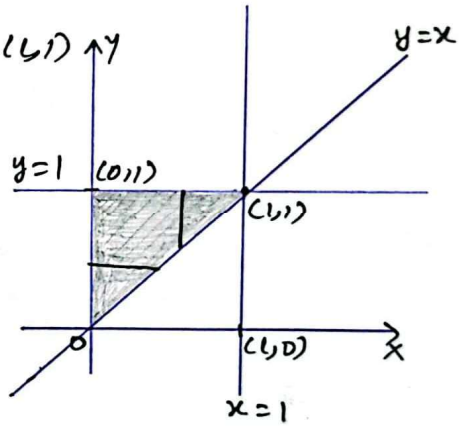
SOLUTION (a)

$$P(0 \le x \le \tfrac{1}{2} \text{ and } \tfrac{1}{4} \le y \le \tfrac{1}{2}) = \int_{\frac{1}{4}}^{\frac{1}{2}} \int_{0}^{\frac{1}{2}} 4xy\,dx\,dy = \frac{3}{64}$$

$$\text{or} \quad = \int_{0}^{\frac{1}{2}} \int_{\frac{1}{4}}^{\frac{1}{2}} 4xy\,dy\,dx = \frac{3}{64}$$

(b) $P(x<y)$

Associated line is $y=x$ $A(0,0)$, $B(1,1)$



we can solve this question both Horizontally and Vertically Movement.

Moving Horizontally

$$\int_0^1 \int_0^y 4xy\,dx\,dy$$

$$= 4 \int_0^1 \left| \frac{x^2}{2} y \right|_0^y dy$$

$$= 2 \int_0^1 y^3 dy = 2 \left| \frac{y^4}{4} \right|_0^1 = \frac{1}{2}(1-0) = \frac{1}{2}$$

Moving Vertically

$$\int_0^1 \int_x^1 4xy\,dy\,dx$$

$$= 4 \int_0^1 \left| x \frac{y^2}{2} \right|_x^1 dx$$

$$= 2 \int_0^1 (x - x^3)\,dx = 2 \left| \frac{x^2}{2} - \frac{x^4}{4} \right|_0^1 = 2\left(\frac{1}{2} - \frac{1}{4}\right) = 2 \times \frac{1}{4} = \frac{1}{2}$$

do we get the same answer.

CamScanner

EXAMPLE ② Given the following joint P.d.f

$$f(x,y) = \begin{cases} \frac{1}{8}(6-x-y), & 0 \le x \le 2,\ 2 \le y \le 4 \\ 0, & \text{otherwise} \end{cases}$$

Calculate

(a) $P(X \le \frac{3}{2}, Y \le \frac{5}{2})$

(b) $P(X+Y < 3)$

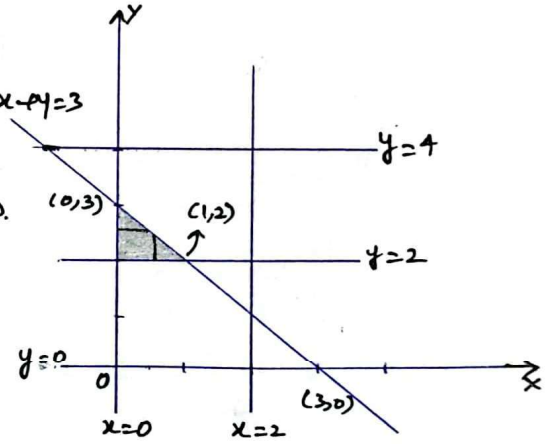
(c) $f(x/y)$ and $f(y/x)$

SOLUTION (a) $P(X \le \frac{3}{2}, Y \le \frac{5}{2}) = \int_0^{3/2}\int_2^{5/2} \frac{1}{8}(6-x-y)\,dy\,dx = \frac{9}{32}$

(b) $P(X+Y<3)$

Associated EQN is $x+y=3$

Points are $A(0,3)$, $B(3,0)$



In this question we can again solve the problem by two Methods.

Moving Horizontally

$$P(X+Y<3) = \frac{1}{8}\int_2^3\int_0^{3-y}(6-x-y)\,dx\,dy$$

$$= \frac{5}{24}$$

Moving vertically

$$P(X+Y<3) = \frac{1}{8}\int_0^1\int_2^{3-x}(6-x-y)\,dy\,dx = \frac{5}{24}$$

(c) Now for $f(x/y)$ and $f(y/x)$ we have to find $g(x)$ and $h(y)$.

$$g(x) = \frac{1}{8}\int_2^4(6-x-y)\,dy = \frac{1}{4}(3-x) \quad \text{Also } h(y) = \frac{1}{8}\int_0^2(6-x-y)\,dx = \frac{1}{4}(5-y)$$

So $f(x/y) = \dfrac{f(x,y)}{h(y)} = \dfrac{1/8(6-x-y)}{\frac{1}{4}(5-y)} = \dfrac{6-x-y}{2(5-y)}$

and $f(y/x) = \dfrac{f(x,y)}{g(x)} = \dfrac{1/8(6-x-y)}{\frac{1}{4}(3-x)} = \dfrac{6-x-y}{2(3-x)}$

EXAMPLE-③ Let X and Y be continuous random variables with joint density function.

$$f(x,y) = \begin{cases} \frac{x}{5} + cy & , \ 0<x<1, \ 1<y<5 \\ 0 & , \ \text{otherwise.} \end{cases}$$

where 'c' is a constant.

(a) Find the value of c.?

(b) Are X and Y independent.

(c) $P(X+Y>3)$.

SOLUTION (a) we know that

$$\int_0^1 \int_1^5 \left(\frac{x}{5} + cy\right) dy\, dx = 1$$

$$\Rightarrow \boxed{c = \frac{1}{20}}$$

(b) $g(x) = \int_1^5 \left(\frac{x}{5} + \frac{1}{20}y\right) dy$ Also $h(y) = \int_0^1 \left(\frac{x}{5} + \frac{1}{20}y\right) dx$

$= \left| \frac{xy}{5} + \frac{y^2}{40} \right|_1^5$ $= \left| \frac{x^2}{10} + \frac{xy}{20} \right|_0^1$

$= \left(\frac{5x}{5} + \frac{25}{40}\right) - \left(\frac{x}{5} + \frac{1}{40}\right)$ $= \frac{1}{10} + \frac{y}{20} - 0$

$g(x) = \frac{4}{5}x + \frac{3}{5} = \frac{4x+3}{5}$ $h(y) = \frac{2+y}{20}$

Now $g(x).h(y) = \frac{4x+3}{5} \times \frac{(y+2)}{20} \neq f(x,y) = \frac{x}{5} + \frac{1}{20}y$

So X and Y are not independent.

(c) $P(X+Y>3)$

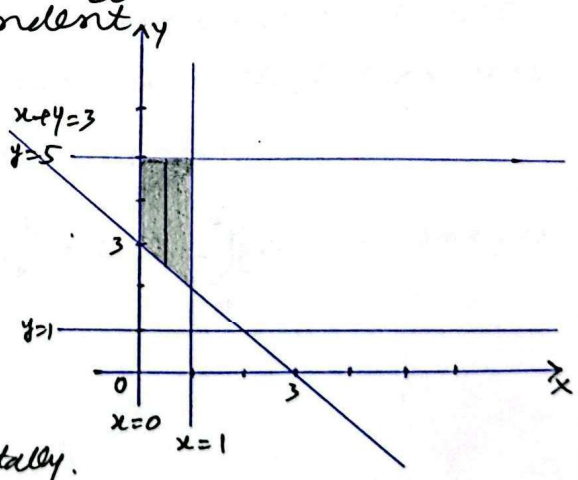
Associated equation is $x+y=3$

$A(0,3)$, $B(3,0)$

Moving vertically (only).

$$P(X+Y>3) = \int_0^1 \int_{3-x}^5 \left(\frac{x}{5} + \frac{y}{20}\right) dy\, dx$$

$$= \frac{11}{15}$$



Note that this question cannot be calculated by moving horizontally.

EXPECTION JOINT CRVS

CamScanner

EXPECTATION OF JOINT CONTINUOUS RANDOM VARIABLES

DEFINITION Let 'x' and 'y' be two continuous Random Variables with a joint density function $f(x,y)$. Then Mean or expected value of random variable $g(x,y)$ is given by

$$E(g(x,y)) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(x,y).f(x,y)\,dx\,dy$$

SOME IMPORTANT FORMULAS AND CONCEPTS

① If x and y are two Random Variables, then

$$E(x \pm y) = E(x) \pm E(y)$$

② If x and y are independent random variables, then

$$E(xy) = E(x).E(y)$$

③ If joint density function of x and y is given by

$$E(x) = \int_{-\infty}^{+\infty} x\,g(x)\,dx$$

and $$E(y) = \int_{-\infty}^{+\infty} y\,h(y)\,dy$$

where $g(x)$ and $h(y)$ are Marginal Functions.

④ $$E(xy) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy\,f(x,y)\,dx\,dy$$

$$E\left(\frac{x}{y}\right) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \frac{x}{y}\,f(x,y)\,dx\,dy$$

⑤ $$E(x+y) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} (x+y)\,f(x,y)\,dx\,dy$$

$$E(x-y) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} (x-y)\,f(x,y)\,dx\,dy$$

CamScanner

EXAMPLE-⑧ The joint density function of the random variables x and y is given by

$$f(x,y) = \begin{cases} 4xy, & 0<x<1,\ 0<y<1 \\ 0, & \text{elsewhere.} \end{cases}$$

Find

(a) E(x) (b) E(y) (c) E(xy) (d) E(x+y)

(e) Are x and y independent random variables.

SOLUTION First we shall find Marginal Prob. fns.

$$g(x) = \int_0^1 4xy\,dy = \left.\frac{4xy^2}{2}\right|_0^1$$

$$= |2xy^2|_0^1$$

$$= 2x - 0 = 2x$$

$$h(y) = \int_0^1 4xy\,dx = 2y$$

Now

(a) E(x)

$$E(x) = \int_0^1 x.g(x)\,dx$$

$$= \int_0^1 x.(2x)\,dx = 2\left|\frac{x^3}{3}\right|_0^1 = \frac{2}{3}(1-0) = \frac{2}{3}$$

(b) E(y)

$$E(y) = \int_0^1 y\,h(y)\,dy$$

$$= \int_0^1 y(2y)\,dy = \frac{2}{3}$$

(c) E(xy)

$$E(xy) = \int_0^1\int_0^1 xy.4xy\,dx\,dy$$

$$= 4\int_0^1\int_0^1 x^2y^2\,dx\,dy$$

$$E(xy) = 4\int_0^1 \left|\frac{x^3}{3}y^2\right|_0^1 dy$$

$$= \frac{4}{3}\int_0^1 (y^2 - 0)\,dy$$

$$= \frac{4}{3}\cdot\left|\frac{y^3}{3}\right|_0^1$$

$$= \frac{4}{9}(1-0) = \frac{4}{9}$$

(d) $$E(x+y) = \int_0^1\int_0^1 (x+y)\,4xy\,dx\,dy$$

$$= 4\int_0^1\int_0^1 (x^2y + xy^2)\,dx\,dy$$

$$= 4\int_0^1 \left|\frac{x^3}{3}y + \frac{x^2}{2}y^2\right|_0^1 dy$$

$$= 4\int_0^1 \left(\frac{y}{3} + \frac{y^2}{2}\right)dy$$

$$= 4\left|\frac{y^2}{6} + \frac{y^3}{6}\right|_0^1$$

$$= 4\left(\frac{1}{6} + \frac{1}{6}\right) = 4\times\frac{2}{6_3} = \frac{4}{3}$$

(e) Independence.

Now $E(xy) = \frac{4}{9}$

$$E(x)\cdot E(y) = \frac{2}{3}\cdot\frac{2}{3} = \frac{4}{9}$$

So $E(xy) = E(x)\cdot E(y)$

$\Rightarrow$ x and y are independent.

CamScanner

EXAMPLE (4) The joint density function of x and y is given by

$$f(x,y) = \begin{cases} \frac{2}{7}(x+2y), & 0<x<1,\ 1<y<2 \\ 0, & \text{elsewhere.} \end{cases}$$

Find

(a) E(x) (b) E(y) (c) E(xy) (d) E(x+y)

(e) Are 'x' and 'y' are independent.

SOLUTION Now first we shall find g(x) and h(y).

$$g(x) = \int_1^2 \frac{2}{7}(x+2y)\,dy$$

$$= \frac{2}{7}\cdot \left| (xy + \not{2}\,\frac{y^2}{\not{2}}) \right|_1^2$$

$$= \frac{2}{7}(2x+4-x-1)$$

$$g(x) = \frac{2}{7}(x+3)$$

$$h(y) = \int_0^1 \frac{2}{7}(x+2y)\,dx$$

$$= \frac{2}{7}\left| \frac{x^2}{2} + 2xy \right|_0^1$$

$$= \frac{2}{7}\left(\frac{1}{2} + 2y\right)$$

$$h(y) = \frac{1}{7}(1+4y) = \frac{1}{7}(4y+1)$$

Now

(a) $$E(x) = \int_0^1 x \cdot \frac{2}{7}(x+3)\,dx$$

$$= \frac{2}{7}\int_0^1 (x^2+3x)\,dx = \frac{2}{7}\left| \frac{x^3}{3} + \frac{3}{2}x^2 \right|_0^1$$

$$= \frac{2}{7}\left(\frac{1}{3} + \frac{3}{2}\right) = \frac{11}{21}$$

(b) $E(Y) = \int_1^2 y \cdot \frac{2}{7}(4y+1)\,dy$

$= \frac{1}{7}\int_1^2 (4y^2+y)\,dy$

$= \frac{1}{7}\left| 4\frac{y^3}{3} + \frac{y^2}{2} \right|_1^2$

$= \frac{1}{7}\left[\left(\frac{32}{3}+2\right) - \left(\frac{4}{3}+\frac{1}{2}\right) \right]$

$= \frac{1}{7}\left[\frac{38}{3} - \frac{11}{6} \right]$

$= \frac{65}{42}$

(c) $E(XY) = \int_1^2 \int_0^1 xy \cdot \frac{2}{7}(x+2y)\,dx\,dy$

$= \frac{2}{7}\int_1^2 \int_0^1 (x^2y + 2xy^2)\,dx\,dy$

$= \frac{2}{7}\int_1^2 \left| \frac{x^3}{3}y + \cancel{2}\cdot\frac{x^2}{\cancel{2}}\cdot y^2 \right|_0^1 dy$

$= \frac{2}{7}\int_1^2 \left(\frac{y}{3} + y^2\right) dy$

$= \frac{2}{7}\left| \frac{y^2}{6} + \frac{y^3}{3} \right|_1^2$

$= \frac{2}{7}\left[\left(\frac{4}{6}+\frac{8}{3}\right) - \left(\frac{1}{6}+\frac{1}{3}\right) \right]$

$= \frac{2}{7}\left[\frac{10}{3} - \frac{1}{2} \right]$

$= \frac{17}{21}$

CamScanner

(d) $E(xy) = \int_{\phi}^{2}\int_{0}^{1} (xy)\frac{2}{7}(x+2y)\,dx\,dy$

$= \frac{2}{7}\int_{\phi}^{2}\int_{0}^{1} (x^2+2xy+xy+2y^2)\,dx\,dy$

$= \frac{2}{7}\int_{1}^{2}\int_{0}^{1} (x^2+3xy+2y^2)\,dx\,dy$

$= \frac{2}{7}\int_{1}^{2} \left| \frac{x^3}{3} + \frac{3}{2}x^2y + 2xy^2 \right|_0^1 dy$

$= \frac{2}{7}\int_{1}^{2} \left(\frac{1}{3} + \frac{3}{2}y + 2y^2\right) dy$

$= \frac{2}{7} \left| \frac{1}{3}y + \frac{3}{2}\cdot\frac{y^2}{2} + \frac{2}{3}y^3 \right|_1^2$

$= \frac{2}{7}\left[\left(\frac{2}{3} + \frac{12}{4} + \frac{16}{3} - \frac{1}{3} - \frac{3}{4} - \frac{2}{3}\right)\right]$

$= \frac{2}{7} \times \frac{29}{4} = \frac{29}{14}$

(e) Independence

Here $E(xy) = \frac{17}{21}$

$E(x) = \frac{11}{21}$

$E(y) = \frac{65}{42}$

$E(x).E(y) = \frac{11}{21} \times \frac{65}{42} = \frac{715}{882}$

$\Rightarrow \quad E(xy) \neq E(x)\,E(y)$

$\Rightarrow$ x and y are not independent.

COVARIANCE AND CORRELATION

COVARIANCE If x and y are random variables with joint probability distribution/density function $f(x,y)$ then covariance of x and y is given by

$$COV(x,y) = \sigma_{xy} = E[(x-E(x))(y-E(y))]$$

or $$COV(x,y) = \sigma_{xy} = E(xy) - E(x)\,E(y)$$

PROPERTIES OF COVARIANCE.

① If x and y are discrete Random Variables then

$$COV(x,y) = \sum_x \sum_y (x-E(x))(y-E(y))$$

② If x and y are continuous Random variables then

$$COV(x,y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x-E(x))(y-E(y))\,dy\,dx$$

③ If x and y are independent Random variables then $COV(x,y) = 0$

④ $COV(x,x) = E[(x-E(x))(x-E(x))]$

$$= E(x-E(x))^2 = Var(x)$$

⑤ If x and y are two random variables then

(a) $Var(x+y) = Var(x) + Var(y) + 2\,COV(x,y)$

(b) $Var(x-y) = Var(x) + Var(y) - 2\,COV(x,y)$

CORRELATION

If x and y are random variables with covariance 'σ_{xy}' and standard deviations σ_x and σ_y respectively. Then correlation coefficient of x and y is given by

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x . \sigma_y}$$

Correlation measures the degree of linear Association between the two random variables. It is not influenced by the units used to measure the variables.

PROPERTIES OF CORRELATION CO-EFFICIENT

① The correlation between two variables is bounded by -1 and 1. ie $-1 \leq \rho_{xy} \leq 1$

② $|\rho_{xy}| = 1$ iff $y = a + bx$. Thus, the correlation between two variables will equal to 1 in absolute value only if there is a perfect linear relationship between the two variables.

③ correlation does not imply causality. Two variables may move together but not be causally related (For eg: shoe size and strength are probably positively related but buying a larger shirt will probably not make you stronger.

EXAMPLE-⑧ If 'x' and 'Y' are CRV's as defined in Exp③. Then find

(a) covariance (σ_{xy})

(b) correlation coefficient (ρ_{xy})

SOLUTION

(a) COVARIANCE

$$\sigma_{xy} = E(XY) - E(X)\,E(Y)$$

$$\sigma_{xy} = \left(\frac{4}{9}\right) - \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)$$

$$\boxed{\sigma_{xy} = 0}$$

(b) CORRELATION

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x . \sigma_y} = \frac{0}{\sigma_x . \sigma_y} = 0$$

$$\Rightarrow \boxed{\rho_{xy} = 0}$$

EXAMPLE-⑨ If 'x' and 'Y' are continuous Random variables as defined in Exp④. Then find.

(a) covariance (σ_{xy})

(b) correlation coefficient (ρ_{xy})

SOLUTION

(a) COVARIANCE

$$\sigma_{xy} = E(XY) - E(X).E(Y)$$

$$= \left(\frac{17}{21}\right) - \left(\frac{11}{21}\right)\left(\frac{65}{42}\right)$$

$$= \frac{17}{21} - \frac{715}{882}$$

$$\sigma_{xy} = -0.0011$$

(b) CORRELATION

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \cdot \sigma_y} \longrightarrow (1)$$

Now $\sigma_x = \sqrt{Var(x)} = \sqrt{E(x^2) - (E(x))^2} \longrightarrow (2)$

So $E(x^2) = \int_0^1 x^2 \cdot \frac{2}{7}(x+3)\,dx$

$$= \frac{2}{7}\int_0^1 (x^3 + 3x^2)\,dx$$

$$= \frac{2}{7}\left| \frac{x^4}{4} + \cancel{3}\cdot\frac{x^3}{\cancel{3}} \right|_0^1$$

$$= \frac{2}{7}\left[\left(\frac{1}{4} + 1\right) - (0+0)\right]$$

$$E(x^2) = \frac{\cancel{2}}{7} \times \frac{5}{\cancel{4}_2} = \frac{5}{14}$$

So $\sigma_x = \sqrt{\left(\frac{5}{14}\right) - \left(\frac{11}{21}\right)^2} = \sqrt{\frac{5}{14} - \frac{121}{441}}$

$$= \frac{73}{882} = 0.0828$$

Also

$$\sigma_y = \sqrt{Var(y)} = \sqrt{E(y^2) - (E(y))^2} \longrightarrow (3)$$

$$E(y^2) = \int_1^2 y^2 \cdot \frac{1}{7}(4y+1)\,dy$$

$$= \frac{1}{7}\int_1^2 (4y^3 + y^2)\,dy$$

$$= \frac{1}{7}\cdot\left| \cancel{4}\frac{y^4}{\cancel{4}} + \frac{y^3}{3} \right|_1^3 = \frac{1}{7}\left[(81+9) - \left(1 + \frac{1}{3}\right)\right]$$

$$E(y^2) = \frac{1}{7}\left[90 - \frac{4}{3}\right] = \frac{1}{7} \times \frac{266}{3} = \frac{38}{3}$$

So

$$\sigma_y = \sqrt{E(y^2) - (E(y))^2}$$

$$= \sqrt{\left(\frac{38}{3}\right) - \left(\frac{65}{42}\right)^2} = \sqrt{\frac{38}{3} - \frac{4225}{1764}} = 3.205$$

Now $\rho_{xy} = \dfrac{\sigma_{xy}}{\sigma_x \cdot \sigma_y} = \dfrac{-0.0011}{(0.0828)(3.205)} = -0.0041$

$$\Rightarrow \boxed{\rho_{xy} = -0.0041}$$

EXAMPLE-⑩ For continuous RV's x and y. Find in Exp(5),

(a) covariance

(b) correlation.

SOLUTION

(a) COVARIANCE

$$\sigma_{xy} = E(xy) - E(x)E(y)$$

$$= \left(\frac{5}{6}\right) - \left(\frac{4}{3}\right)\left(\frac{5}{8_2}\right)$$

$$\sigma_{xy} = \frac{5}{6} - \frac{5}{6} = 0 \quad , \boxed{\sigma_{xy} = 0}$$

(b) CORRELATION

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \cdot \sigma_y}$$

$$\rho_{xy} = \frac{0}{\sigma_x \cdot \sigma_y} = 0$$

$$\Rightarrow \boxed{\rho_{xy} = 0}$$

CamScanner